ENGINEERING MATHEMATICS - 1

Complete Syllabus Master Study Guide & Exam Companion (v6)

Course Codes: EM201T / NEM201T / EMM103T — B.Tech Semester I

Abstract

This master study guide delivers an ultra-detailed, beginner-friendly, topic-by-topic breakdown of the entire 5-Unit Engineering Mathematics-1 syllabus. Every concept is explained in plain, intuitive language with step-by-step problem-solving algorithms, clear visual analogies, and calculator tricks. Special emphasis is placed on every single Past Year Question (PYQ) from the Mid-Semester (2024, 2025, 2026), Semester End Examinations (SEE 2024, 2025), EVEN Re-Examinations (April 2026), and Mid-Sem Coursework Assignments. Topics that have appeared in exams are highlighted in gold as High-Yield PYQ Topics, while non-PYQ syllabus topics are fully explained with custom worked illustration problems to guarantee 100% exam readiness.

Document Features & Navigation

- **100% Syllabus Coverage**: Every topic from Units I through V of the official syllabus is explained from scratch.
- **Gold High-Yield PYQ Callouts**: Every recurring pattern from 2024–2026 exam papers is explicitly tagged with exact mark allocations.
- **Blue Concept & Illustration Callouts**: Non-PYQ syllabus topics feature custom, fully solved numerical illustrations.
- **Ultra-Simple Step-by-Step Recipes**: Complex proofs and calculations are broken down into numbered 1-2-3 action steps.
- **CASIO fx-991ES PLUS Calculator Companion**: Practical button steps for checking limits, derivatives, and vectors.

Institute of Advanced Research — School of Engineering
Generated for Student Review & Semester End Examination Prep

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1 Unit I: Partial Differentiation & Multivariable Calculus

1.1 Functions of Several Variables, 3D Geometry & Surfaces

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Functions of Two or More Variables

What is a Multivariable Function?
In single-variable calculus, you studied $y = f(x)$, where one input x determines one output y. In multivariable calculus, a function $z = f(x, y)$ takes **two independent inputs** (x and y) and gives **one output** (z).

- **Visual Analogy**: Imagine a terrain map. Your position on the ground is (x, y), and the altitude (height of the hill) at that exact spot is $z = f(x, y)$.
- **3D Surface**: While $y = f(x)$ draws a 2D curve on a flat plane, $z = f(x, y)$ draws a **3D surface** floating in space.
- **Domain & Range**: The domain is a 2D region in the xy-plane where (x, y) points live. The range is the set of all resulting z-values (heights).

Example Illustration: Consider $f(x, y) = x^2 + y^2$.

- At the origin $(0, 0)$, $f(0, 0) = 0^2 + 0^2 = 0$.
- At point $(1, 2)$, $f(1, 2) = 1^2 + 2^2 = 5$.
- Graphically, this forms a 3D bowl shape known as a *paraboloid*.

1.2 Limits & Continuity in Higher Dimensions

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Limits in 3D Space The Path Non-Existence Test

The Key Difference Between 2D and 3D Limits: In 2D calculus ($\lim_{x\to a} f(x)$), you could approach the point a from only **two directions**: left ($x \to a^-$) or right ($x \to a^+$). In 3D calculus ($\lim_{(x,y)\to(a,b)} f(x,y)$), you can approach the point (a,b) along **infinitely many paths** in the 2D plane (straight lines, parabolas, spirals)!
Definition of Limit Existence: A multivariable limit exists and equals L if and only if $f(x,y)$ approaches L along **EVERY single path** approaching (a,b).
Algorithm: How to Prove a 3D Limit Does NOT Exist (Path Test):

1. Substitute $y = mx$ (straight line paths passing through the origin $(0,0)$) into the limit expression.
2. Simplify the expression. If m remains in the final answer, the limit **depends on the slope** m.
3. Since different paths (different values of m) yield different limits, conclude: **The limit does not exist!**

Illustrative Worked Problem: Evaluate $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$.

Solution Step-by-Step. 1. Let $y = mx$. Substitute $y = mx$ into the limit:

$$\lim_{x\to 0} \frac{x(mx)}{x^2+(mx)^2} = \lim_{x\to 0} \frac{mx^2}{x^2(1+m^2)} = \frac{m}{1+m^2}$$

2. Along the line $y = x$ ($m = 1$), the limit is $\frac{1}{1+1} = \frac{1}{2}$. 3. Along the line $y = 2x$ ($m = 2$), the limit is $\frac{2}{1+4} = \frac{2}{5}$. 4. Since $\frac{1}{2} \neq \frac{2}{5}$, the limit depends on the path m. Therefore, the limit **does not exist**. □

IMPORTANT PYQ / EXAM TOPIC: Piecewise Boundary Continuity Solver (2-Mark Core Pattern)

Exam Context: Appears repeatedly in Mid-Sem 2024, Mid-Sem 2025, and EVEN RE-EXAM April 2026! **Rule for Continuity**: A piecewise function $f(x)$ is continuous at a boundary point $x = c$ if:

$$\lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = f(c)$$

Solved PYQ 1 (Mid-Sem 2024, Mid-Sem 2025): Consider the function $f(x) = \begin{cases} A+x & \text{if } x < 2 \\ 1+x^2 & \text{if } x \geq 2 \end{cases}$. Find the value of A such that $f(x)$ is continuous at $x = 2$.

Step-by-Step Solution. 1. Calculate Left-Hand Limit (LHL) as $x \to 2^-$:

$$\text{LHL} = \lim_{x\to 2^-} (A+x) = A+2$$

2. Calculate Right-Hand Limit (RHL) / Function Value $f(2)$ as $x \to 2^+$:

$$\text{RHL} = f(2) = 1+(2)^2 = 1+4 = 5$$

3. Equate LHL = RHL for continuity:

$$A+2 = 5 \implies \mathbf{A = 3}$$

□

Solved PYQ 2 (EVEN RE-EXAM April 2026): Consider $f(x) = \begin{cases} 2x^2+3 & \text{if } x \leq 2 \\ 3x+B & \text{if } x > 2 \end{cases}$. Find B for continuity at $x = 2$.

Step-by-Step Solution. 1. LHL at $x = 2$: $f(2) = 2(2)^2 + 3 = 8 + 3 = 11$. 2. RHL at $x = 2$: $\lim_{x\to 2^+}(3x + B) = 3(2) + B = 6 + B$. 3. Equate LHL = RHL: $6 + B = 11 \implies \mathbf{B = 5}$. □

1.3 Partial Derivatives & Clairaut's Theorem

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: What is a Partial Derivative?

When differentiating $z = f(x, y)$:

- **Partial w.r.t. x ($\frac{\partial f}{\partial x}$ or f_x)**: Treat x as the variable and **hold y completely constant** (like a number like 5).
- **Partial w.r.t. y ($\frac{\partial f}{\partial y}$ or f_y)**: Treat y as the variable and **hold x completely constant**.

IMPORTANT PYQ / EXAM TOPIC: First & Second-Order Partial Derivatives (Mid-Sem 2024, 2025, 2026, SEE 2024, 2025)

Solved PYQ 1 (Repeated in 5 Exam Papers!): If $f(x, y) = x^2 + y^2 + x^2y^2$, find all first-order partial derivatives.

Step-by-Step Solution. 1. Find $\frac{\partial f}{\partial x}$ (treat y as constant):

$$\frac{\partial f}{\partial x} = \frac{d}{dx}(x^2) + 0 + y^2\frac{d}{dx}(x^2) = \mathbf{2x + 2xy^2}$$

2. Find $\frac{\partial f}{\partial y}$ (treat x as constant):

$$\frac{\partial f}{\partial y} = 0 + \frac{d}{dy}(y^2) + x^2\frac{d}{dy}(y^2) = \mathbf{2y + 2x^2y}$$

□

Solved PYQ 2 (SEE 2024, SEE 2025): Find all partial derivatives of $f(x, y) = 6x^2+13xy+4y^2$.

Step-by-Step Solution. 1. $\frac{\partial f}{\partial x} = \frac{d}{dx}(6x^2) + 13y\frac{d}{dx}(x) + 0 = \mathbf{12x + 13y}$. 2. $\frac{\partial f}{\partial y} = 0 + 13x\frac{d}{dy}(y) + \frac{d}{dy}(4y^2) = \mathbf{13x + 8y}$. □

Clairaut's Mixed Partials Equality Theorem: If $f(x, y)$ and its partial derivatives are continuous, then the order of mixed partial differentiation does not matter:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} \quad \text{or} \quad f_{xy} = f_{yx}$$

1.4 Multivariable Chain Rule & Total Differentials

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Multivariable Chain Rule Rules

Case 1: $z = f(x, y)$ where $x = x(t)$ and $y = y(t)$ (Single Parameter t):

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

Case 2: $z = f(x, y)$ **where** $x = u + v$ **and** $y = u - v$ **(Two Parameters** u, v**):**

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u} = \frac{\partial f}{\partial x}(1) + \frac{\partial f}{\partial y}(1) = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} + \frac{\partial \mathbf{f}}{\partial \mathbf{y}}$$
$$\frac{\partial z}{\partial v} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial v} = \frac{\partial f}{\partial x}(1) + \frac{\partial f}{\partial y}(-1) = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} - \frac{\partial \mathbf{f}}{\partial \mathbf{y}}$$

(Note: MCQ Q4 in Mid-Sem 2025 and Mid-Sem 2026 tests this exact formula!)

IMPORTANT PYQ / EXAM TOPIC: Total Differential Coefficient Under Constraint (Mid-Sem 2025, Mid-Sem 2026)

Solved PYQ: Find the total differential coefficient of $u = x^2 y$ with respect to x where $x^2 + xy + y^2 = 1$.

Step-by-Step Solution. 1. Formula for Total Differential Coefficient $\frac{du}{dx}$:

$$\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\frac{dy}{dx}$$

2. Compute partials of $u = x^2 y$: $\frac{\partial u}{\partial x} = 2xy, \quad \frac{\partial u}{\partial y} = x^2$. 3. Find $\frac{dy}{dx}$ from constraint $x^2 + xy + y^2 = 1$ via implicit differentiation: Differentiate w.r.t x:

$$2x + \left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0 \implies (x + 2y)\frac{dy}{dx} = -(2x + y) \implies \frac{dy}{dx} = -\frac{2x + y}{x + 2y}$$

4. Substitute into total differential formula:

$$\frac{\mathbf{du}}{\mathbf{dx}} = \mathbf{2xy} - \mathbf{x^2}\left(\frac{\mathbf{2x + y}}{\mathbf{x + 2y}}\right) = \frac{\mathbf{2xy(x + 2y) - x^2(2x + y)}}{\mathbf{x + 2y}} = \frac{\mathbf{3x^2y + 4xy^2 - 2x^3}}{\mathbf{x + 2y}}$$

□

1.5 Homogeneous Functions & Euler's Theorem

IMPORTANT PYQ / EXAM TOPIC: Euler's Theorem on Homogeneous Functions (Mid-Sem 2024, 2025, 2026, Assignment Q27)

Definition of Homogeneous Function: A function $f(x, y)$ is homogeneous of degree n if substituting $x \to tx$ and $y \to ty$ yields:

$$f(tx, ty) = t^n f(x, y)$$

Euler's Theorem (First Order Statement & Proof): If $u = f(x, y)$ is a homogeneous function of degree n, then:

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$$

Complete Proof Recipe (5-Mark Exam Proof):

Proof. 1. Since u is homogeneous of degree n, we can write $u = x^n F\left(\frac{y}{x}\right)$. 2. Differentiate u partially w.r.t x using Product Rule:

$$\frac{\partial u}{\partial x} = nx^{n-1}F\left(\frac{y}{x}\right) + x^n F'\left(\frac{y}{x}\right)\left(-\frac{y}{x^2}\right) = nx^{n-1}F\left(\frac{y}{x}\right) - yx^{n-2}F'\left(\frac{y}{x}\right)$$

Multiply by x:

$$x\frac{\partial u}{\partial x} = nx^n F\left(\frac{y}{x}\right) - xyx^{n-2}F'\left(\frac{y}{x}\right) = nu - yx^{n-1}F'\left(\frac{y}{x}\right) \quad \text{— (Eq 1)}$$

3. Differentiate u partially w.r.t y:

$$\frac{\partial u}{\partial y} = x^n F'\left(\frac{y}{x}\right)\left(\frac{1}{x}\right) = x^{n-1}F'\left(\frac{y}{x}\right)$$

Multiply by y:

$$y\frac{\partial u}{\partial y} = yx^{n-1}F'\left(\frac{y}{x}\right) \quad \text{— (Eq 2)}$$

4. Add (Eq 1) and (Eq 2):

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu - yx^{n-1}F'\left(\frac{y}{x}\right) + yx^{n-1}F'\left(\frac{y}{x}\right) = \mathbf{nu} \quad \text{(Hence Proved!)}$$

□

1.6 Gradient Vectors, Directional Derivatives & Extrema

IMPORTANT PYQ / EXAM TOPIC: Gradient Vector $\nabla\Phi$ at a 3D Point (SEE 2024, SEE 2025, RE-EXAM 2026)

Definition: For a scalar field $\Phi(x, y, z)$, the gradient vector is:

$$\nabla\Phi = \text{grad }\Phi = \frac{\partial\Phi}{\partial x}\mathbf{i} + \frac{\partial\Phi}{\partial y}\mathbf{j} + \frac{\partial\Phi}{\partial z}\mathbf{k}$$

Solved PYQ (Repeated in SEE 2024, 2025, RE-EXAM 2026): Find $\nabla\Phi$ for $\Phi(x, y, z) = 3x^2y - y^3z^2$ at the point $(1, -2, -1)$.

Step-by-Step Solution. 1. Compute partial derivatives:

$$\frac{\partial\Phi}{\partial x} = 6xy$$
$$\frac{\partial\Phi}{\partial y} = 3x^2 - 3y^2z^2$$
$$\frac{\partial\Phi}{\partial z} = -2y^3z$$

2. Evaluate each partial at point $(x = 1, y = -2, z = -1)$:

$$\left.\frac{\partial\Phi}{\partial x}\right|_{(1,-2,-1)} = 6(1)(-2) = -12$$
$$\left.\frac{\partial\Phi}{\partial y}\right|_{(1,-2,-1)} = 3(1)^2 - 3(-2)^2(-1)^2 = 3 - 3(4)(1) = 3 - 12 = -9$$
$$\left.\frac{\partial\Phi}{\partial z}\right|_{(1,-2,-1)} = -2(-2)^3(-1) = -2(-8)(-1) = -16$$

3. Combine into gradient vector:

$$\nabla\mathbf{\Phi}(\mathbf{1}, \mathbf{-2}, \mathbf{-1}) = \mathbf{-12i - 9j - 16k}$$

□

IMPORTANT PYQ / EXAM TOPIC: Directional Derivatives (Mid-Sem 2025, Mid-Sem 2026)

Formula: The directional derivative of $f(x, y, z)$ along a direction vector $\mathbf{v}$ is:

$$D_{\mathbf{u}} f = \nabla f \cdot \hat{\mathbf{u}} \quad \text{where } \hat{\mathbf{u}} = \frac{\mathbf{v}}{|\mathbf{v}|} \text{ is the unit vector.}$$

Solved PYQ (Mid-Sem 2025, Mid-Sem 2026): Find the directional derivative of $f(x, y, z) = xyz$ in the direction $3\mathbf{i} - 4\mathbf{k}$ at the point $(1, -1, 1)$.

Step-by-Step Solution. 1. Find gradient ∇f: $\nabla f = yz\,\mathbf{i} + xz\,\mathbf{j} + xy\,\mathbf{k}$. 2. Evaluate ∇f at $(1, -1, 1)$: $\nabla f(1, -1, 1) = (-1)(1)\,\mathbf{i} + (1)(1)\,\mathbf{j} + (1)(-1)\,\mathbf{k} = -\mathbf{i} + \mathbf{j} - \mathbf{k}$. 3. Find unit vector $\hat{\mathbf{u}}$ for direction $\mathbf{v} = 3\mathbf{i} - 4\mathbf{k}$: $|\mathbf{v}| = \sqrt{3^2 + 0^2 + (-4)^2} = \sqrt{9 + 16} = 5$. $\hat{\mathbf{u}} = \frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{k}$. 4. Compute Dot Product $\nabla f \cdot \hat{\mathbf{u}}$:

$$D_{\mathbf{u}} f = (-1)\left(\frac{3}{5}\right) + (1)(0) + (-1)\left(-\frac{4}{5}\right) = -\frac{3}{5} + \frac{4}{5} = \frac{\mathbf{1}}{\mathbf{5}} = \mathbf{0.2}$$

□

IMPORTANT PYQ / EXAM TOPIC: Relative Extrema of 2-Variable Functions (SEE 2024, SEE 2025, RE-EXAM 2026)

3-Step Classification Algorithm: 1. Find critical points by setting $f_x = 0$ and $f_y = 0$. 2. Calculate second partials: $A = f_{xx}$, $B = f_{xy}$, $C = f_{yy}$. 3. Evaluate discriminant $D = AC - B^2$:

- If $D > 0$ and $A > 0 \implies$ **Relative Minimum**.
- If $D > 0$ and $A < 0 \implies$ **Relative Maximum**.
- If $D < 0 \implies$ **Saddle Point** (neither max nor min).
- If $D = 0 \implies$ **Test Inconclusive**.

Solved PYQ (SEE 2024, SEE 2025, RE-EXAM 2026): Determine the relative extrema of $f(x, y) = 6x^2 + 10y^2 + 2x - 3y + 15$.

Step-by-Step Solution. 1. Find critical points: $f_x = 12x + 2 = 0 \implies x = -2/12 = -1/6$. $f_y = 20y - 3 = 0 \implies y = 3/20$. Critical point is $\left(-\frac{1}{6}, \frac{3}{20}\right)$. 2. Compute second partials: $A = f_{xx} = 12$, $B = f_{xy} = 0$, $C = f_{yy} = 20$. 3. Evaluate Discriminant D: $D = AC - B^2 = (12)(20) - 0^2 = 240$. 4. Since $D = 240 > 0$ and $A = 12 > 0$, the point $\left(-\frac{1}{6}, \frac{3}{20}\right)$ is a **Relative Minimum**. □

2 Unit II: Infinite Series & Expansions

2.1 Sequences & Infinite Series Basics

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: What is an Infinite Series?

An infinite series is the sum of infinitely many terms of a sequence:

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots + a_n + \ldots$$

Convergence vs Divergence:

- **Convergent Series**: If the sum of terms approaches a fixed finite number S as $n \to \infty$, the series *converges*.
- **Divergent Series**: If the sum grows to $+\infty$, $-\infty$, or oscillates without settling on a single number, the series *diverges*.

2.2 Master Summary of Series Convergence Tests

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Summary Table of Convergence Tests

Test Name	Mathematical Condition	Conclusion / Rule
nth-Term Test	$\lim_{n\to\infty} a_n \neq 0$	Diverges immediately! (If limit is 0, test is inconclusive).
Geometric Series	$\sum_{n=0}^{\infty} ar^n$	Converges if $\|r\| < 1$; Diverges if $\|r\| \geq 1$.
p-Series Test	$\sum_{n=1}^{\infty} \frac{1}{n^p}$	Converges if $p > 1$; Diverges if $p \leq 1$.
D'Alembert's Ratio Test	$L = \lim_{n\to\infty} \left\|\frac{a_{n+1}}{a_n}\right\|$	Converges if $L < 1$; Diverges if $L > 1$; Inconclusive if $L = 1$.
Cauchy's Root Test	$L = \lim_{n\to\infty} (a_n)^{1/n}$	Converges if $L < 1$; Diverges if $L > 1$; Inconclusive if $L = 1$.
Leibniz Alternating Test	$\sum(-1)^{n-1} b_n$ with $b_{n+1} \leq b_n$	Converges if $\lim_{n\to\infty} b_n = 0$.

IMPORTANT PYQ / EXAM TOPIC: Series Summation & Ratio Test (SEE 2024, SEE 2025)

Solved PYQ (SEE 2024, SEE 2025): Evaluate $\sum_{i=1}^{4}(2n+3)$.

Step-by-Step Solution. 1. Write out the 4 terms for $n = 1, 2, 3, 4$: For $n = 1$: $2(1) + 3 = 5$
For $n = 2$: $2(2) + 3 = 7$
For $n = 3$: $2(3) + 3 = 9$
For $n = 4$: $2(4) + 3 = 11$ 2. Sum the terms: $5 + 7 + 9 + 11 = \mathbf{32}$. □

2.3 Taylor's & Maclaurin's Series

IMPORTANT PYQ / EXAM TOPIC: Maclaurin Series Derivation & Standard Expansions (Mid-Sem 2026)

Formula: The Maclaurin series represents a function $f(x)$ expanded around $x = 0$:

$$f(x) = f(0) + x \cdot f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \cdots + \frac{x^n}{n!} f^{(n)}(0) + \ldots$$

Solved PYQ (Mid-Sem 2026): Determine the Taylor series expansion at $x = 0$ for $f(x) = e^x$.

Step-by-Step Solution. 1. Find derivatives of $f(x) = e^x$: $f(x) = e^x \implies f(0) = e^0 = 1$
$f'(x) = e^x \implies f'(0) = 1$
$f''(x) = e^x \implies f''(0) = 1$
$f'''(x) = e^x \implies f'''(0) = 1$ 2. Substitute values into Maclaurin expansion:

$$e^x = 1 + x(1) + \frac{x^2}{2!}(1) + \frac{x^3}{3!}(1) + \cdots = \mathbf{1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}}$$

□

Standard Maclaurin Expansions Summary Table:

Function	Series Expansion	Interval of Conv.
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	$(-\infty, \infty)$
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$	$(-\infty, \infty)$
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$	$(-\infty, \infty)$
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$	$(-1, 1]$
$\frac{1}{1-x}$	$1 + x + x^2 + x^3 + \dots$	$(-1, 1)$

3 Unit III: Improper Integrals & Applications of Revolution

3.1 Improper Integrals (Type 1 & Type 2)

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: What makes an Integral "Improper"?

A definite integral $\int_a^b f(x)\,dx$ is improper if:

- **Type 1 (Infinite Limits)**: One or both limits are infinite (∞ or $-\infty$). Example: $\int_1^\infty \frac{1}{x^2}dx = \lim_{t\to\infty}\left[-\frac{1}{x}\right]_1^t = \lim_{t\to\infty}\left(1 - \frac{1}{t}\right) = \mathbf{1}$ (Convergent).
- **Type 2 (Unbounded Integrand)**: The integrand $f(x)$ goes to $\pm\infty$ at some point in $[a, b]$. Example: $\int_0^1 \frac{1}{\sqrt{x}}dx = \lim_{t\to 0^+}[2\sqrt{x}]_t^1 = 2(1-0) = \mathbf{2}$ (Convergent).

3.2 Beta & Gamma Functions

IMPORTANT PYQ / EXAM TOPIC: Beta & Gamma Functions & Proofs (EVEN RE-EXAM April 2026)

Definitions:

$$\text{Gamma Function: } \Gamma(n) = \int_0^\infty x^{n-1}e^{-x}\,dx \quad (n > 0)$$

$$\text{Beta Function: } \beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx \quad (m > 0, n > 0)$$

Essential Properties & Formulas: 1. Recurrence Relation: $\Gamma(n+1) = n\Gamma(n) = n!$ (for integer n). 2. Half-Integer Value: $\mathbf{\Gamma(1/2) = \sqrt{\pi}}$. 3. Beta-Gamma Conversion Identity: $\beta(\mathbf{m,n}) = \frac{\mathbf{\Gamma(m)\Gamma(n)}}{\mathbf{\Gamma(m+n)}}$.
Solved PYQ 1 (RE-EXAM 2026): Prove that $\beta(m,n) = \beta(n,m)$ (Symmetry Property).

Step-by-Step Solution. 1. Start with definition: $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$. 2. Use substitution $u = 1 - x \implies x = 1 - u$, and $dx = -du$. 3. Change limits: when $x = 0 \to u = 1$; when

$x = 1 \to u = 0$. 4. Substitute:

$$\beta(m,n) = \int_1^0 (1-u)^{m-1}u^{n-1}(-du) = \int_0^1 u^{n-1}(1-u)^{m-1}\,du = \beta(\mathbf{n},\mathbf{m}) \quad \text{(Proved!)}$$

□

Solved PYQ 2 (RE-EXAM 2026): Find $\Gamma(5/2)$.

Step-by-Step Solution. Apply $\Gamma(n+1) = n\Gamma(n)$ repeatedly:

$$\Gamma(5/2) = \Gamma(3/2+1) = \frac{3}{2}\Gamma(3/2) = \frac{3}{2}\cdot\frac{1}{2}\Gamma(1/2) = \frac{3}{4}\sqrt{\pi} = \boxed{\frac{3\sqrt{\pi}}{4}}$$

□

Solved PYQ 3 (RE-EXAM 2026): If $\Gamma(1.8) = 0.9314$, what is $\Gamma(-2.2)$?

Step-by-Step Solution. Use recurrence formula backwards: $\Gamma(n) = \frac{\Gamma(n+1)}{n}$. 1. $\Gamma(-1.2) = \frac{\Gamma(-0.2)}{-1.2}$ 2. $\Gamma(-0.2) = \frac{\Gamma(0.8)}{-0.2}$ 3. $\Gamma(0.8) = \frac{\Gamma(1.8)}{0.8} = \frac{0.9314}{0.8} = 1.16425$. 4. $\Gamma(-0.2) = \frac{1.16425}{-0.2} = -5.82125$. 5. $\Gamma(-1.2) = \frac{-5.82125}{-1.2} = 4.85104$. 6. $\Gamma(-2.2) = \frac{4.85104}{-2.2} = \mathbf{-2.20502}$. □

3.3 Applications of Integration: Area & Volume of Revolution

IMPORTANT PYQ / EXAM TOPIC: Volume of Solids of Revolution (SEE 2024, SEE 2025, RE-EXAM 2026)

Volume Formulas:

- Revolved about x-axis (Disk Method): $V = \pi \int_a^b [y(x)]^2\,dx$.
- Revolved about y-axis (Cylindrical Shell Method): $V = 2\pi \int_a^b x \cdot y(x)\,dx$.

Solved PYQ (SEE 2024, 2025, RE-EXAM 2026): Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and the x-axis over $[1,3]$ about the y-axis.

Step-by-Step Solution. 1. Identify integration bounds: $x \in [1,3]$, and height function $y = x^2$. 2. Apply Cylindrical Shell Formula about y-axis:

$$V = 2\pi \int_1^3 x \cdot y\,dx = 2\pi \int_1^3 x(x^2)\,dx = 2\pi \int_1^3 x^3\,dx$$

3. Integrate x^3:

$$V = 2\pi \left[\frac{x^4}{4}\right]_1^3 = 2\pi\left(\frac{81}{4} - \frac{1}{4}\right) = 2\pi\left(\frac{80}{4}\right) = 2\pi(20) = \mathbf{40\pi \approx 125.66}$$

□

4 Unit IV: Multiple Integration & Vector Calculus

4.1 Double & Triple Integrals in Cartesian Coordinates

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: How to Evaluate Double Triple Integrals

The Golden Rule: Work from the **INNMOST integral outwards**!

- Evaluate the innermost integral while treating all other variables as constant.
- Substitute the inner limits, then move to the next outer variable.

IMPORTANT PYQ / EXAM TOPIC: Double Triple Integral Evaluation (SEE 2024, SEE 2025, 10-Mark Section C)

Solved PYQ 1 (5-Marker - SEE 2024, SEE 2025): Evaluate $\int_0^1 \int_0^2 y^2 x \, dy \, dx$.

Step-by-Step Solution. 1. Inner integral w.r.t y from 0 to 2 (treat x as constant):

$$\int_0^2 y^2 x \, dy = x \left[\frac{y^3}{3}\right]_0^2 = x\left(\frac{8}{3} - 0\right) = \frac{8}{3}x$$

2. Outer integral w.r.t x from 0 to 1:

$$\int_0^1 \frac{8}{3}x \, dx = \frac{8}{3}\left[\frac{x^2}{2}\right]_0^1 = \frac{8}{3}\left(\frac{1}{2}\right) = \mathbf{\frac{4}{3}}$$

□

Solved PYQ 2 (10-Marker - SEE 2024, SEE 2025): Evaluate 3D Triple Integral $\int_0^2 \int_1^3 \int_1^2 xy^2 z \, dz \, dy \, dx$.

Step-by-Step Solution. 1. Innermost integral w.r.t z from 1 to 2:

$$\int_1^2 xy^2 z \, dz = xy^2 \left[\frac{z^2}{2}\right]_1^2 = xy^2\left(\frac{4}{2} - \frac{1}{2}\right) = \frac{3}{2}xy^2$$

2. Middle integral w.r.t y from 1 to 3:

$$\int_1^3 \frac{3}{2}xy^2 \, dy = \frac{3}{2}x\left[\frac{y^3}{3}\right]_1^3 = \frac{1}{2}x(27-1) = \frac{1}{2}x(26) = 13x$$

3. Outermost integral w.r.t x from 0 to 2:

$$\int_0^2 13x \, dx = 13\left[\frac{x^2}{2}\right]_0^2 = 13\left(\frac{4}{2}\right) = 13(2) = \mathbf{26}$$

□

4.2 Cartesian to Polar Transformations & Change of Variables

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Polar Coordinates Transformation ($x, y \to r, \theta$)

Conversion Formulas:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad x^2 + y^2 = r^2$$

Area Element Transformation (Jacobian $J = r$):

$$dx\,dy = r\,dr\,d\theta$$

IMPORTANT PYQ / EXAM TOPIC: 10-Mark Area by Double Integration (SEE 2024, SEE 2025, RE-EXAM 2026)

Solved PYQ 1 (Area of Ellipse - 10 Marks): Find by double integration the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Step-by-Step Solution. 1. Use symmetry: Total Area $A = 4 \times$ (Area in First Quadrant). 2. Set up limits for 1st quadrant: $x \in [0, a]$, $y \in \left[0, b\sqrt{1 - \frac{x^2}{a^2}}\right]$.

$$A = 4\int_0^a \left(\int_0^{b\sqrt{1-x^2/a^2}} dy\right) dx = 4\int_0^a b\sqrt{1 - \frac{x^2}{a^2}}\,dx = \frac{4b}{a}\int_0^a \sqrt{a^2 - x^2}\,dx$$

3. Use standard integral $\int \sqrt{a^2 - x^2}dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\arcsin\left(\frac{x}{a}\right)$:

$$A = \frac{4b}{a}\left[0 + \frac{a^2}{2}\left(\frac{\pi}{2}\right)\right] = \frac{4b}{a}\left(\frac{\pi a^2}{4}\right) = \boldsymbol{\pi ab}$$

□

Solved PYQ 2 (Area Between Curves - 10 Marks): Find the area between parabola $y = 4x - x^2$ and line $y = x$.

Step-by-Step Solution. 1. Intersection points: $4x - x^2 = x \implies x^2 - 3x = 0 \implies x = 0, x = 3$. 2. Double integral for area:

$$A = \int_0^3 \left(\int_x^{4x-x^2} dy\right) dx = \int_0^3 (3x - x^2)\,dx = \left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_0^3 = \frac{27}{2} - 9 = \boldsymbol{\frac{9}{2} = 4.5}$$

□

Solved PYQ 3 (Polar Circles Area - 10 Marks - SEE 2024, SEE 2025, RE-EXAM 2026): Find the area bounded between the circles $r = 2\sin\theta$ and $r = 4\sin\theta$.

Step-by-Step Solution. 1. Setup polar area double integral: $A = \int_0^{\pi} \int_{2\sin\theta}^{4\sin\theta} r\,dr\,d\theta$. 2. Inner integral w.r.t r:

$$\left[\frac{r^2}{2}\right]_{2\sin\theta}^{4\sin\theta} = \frac{16\sin^2\theta - 4\sin^2\theta}{2} = 6\sin^2\theta = 6\left(\frac{1 - \cos 2\theta}{2}\right) = 3(1 - \cos 2\theta)$$

3. Outer integral w.r.t θ from 0 to π:

$$A = \int_0^{\pi} 3(1 - \cos 2\theta)\,d\theta = 3\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi} = 3(\pi - 0) = \boldsymbol{3\pi \approx 9.42}$$

□

4.3 Vector Calculus: Divergence, Curl & Integral Theorems

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Divergence Curl Definitions

For a vector field $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$:

- **Divergence ($\nabla \cdot \mathbf{F}$)**: Dot product of del operator with $\mathbf{F}$ (yields a *scalar*).

$$\nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

If $\nabla \cdot \mathbf{F} = 0 \implies \mathbf{F}$ is **Solenoidal**.

- **Curl ($\nabla \times \mathbf{F}$)**: Cross product of del operator with $\mathbf{F}$ (yields a *vector*).

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

If $\nabla \times \mathbf{F} = \mathbf{0} \implies \mathbf{F}$ is **Irrotational**.

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Summary of Classical Integral Theorems

1. **Green's Theorem in the Plane**: Converts a line integral around a closed 2D curve C into a double integral over region R:

$$\oint_C (P\,dx + Q\,dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

2. **Stokes' Theorem**: Relates a 3D line integral around C to a surface integral of Curl $\mathbf{F}$:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dS$$

3. **Gauss's Divergence Theorem**: Relates a surface integral over a closed 3D surface S to a volume integral of Divergence $\mathbf{F}$:

$$\iint_S \mathbf{F} \cdot \mathbf{n}\, dS = \iiint_V (\nabla \cdot \mathbf{F})\, dV$$

5 Unit V: Ordinary & Partial Differential Equations

5.1 First-Order Linear Differential Equations (Integrating Factors)

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Solving First-Order LDEs

Standard Form: $\frac{dy}{dx} + P(x)y = Q(x)$. **3-Step Solution Recipe**: 1. Identify $P(x)$ and $Q(x)$. 2. Calculate Integrating Factor (IF): IF $= e^{\int P(x)dx}$. 3. Write General Solution: $y \cdot (\text{IF}) = \int Q(x) \cdot (\text{IF})\, dx + C$.
Worked Example: Solve $\frac{dy}{dx} + \frac{2}{x}y = x^3$.

Solution. 1. $P(x) = \frac{2}{x}$, $Q(x) = x^3$. 2. IF $= e^{\int \frac{2}{x}dx} = e^{2\ln x} = e^{\ln(x^2)} = x^2$. 3. General Solution: $y \cdot x^2 = \int x^3 \cdot x^2\, dx + C = \int x^5 dx + C = \frac{x^6}{6} + C \implies \mathbf{y = \frac{x^4}{6} + \frac{C}{x^2}}$. □

5.2 Second-Order Linear ODEs with Constant Coefficients

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Complementary Function (CF) Particular Integral (PI)

General Solution: $y = y_{\text{CF}} + y_{\text{PI}}$.
Auxiliary Equation Roots for y_{CF}: For equation $ay'' + by' + cy = 0 \implies am^2 + bm + c = 0$:

- **Real & Distinct Roots** ($m_1 \neq m_2$): $y_{\text{CF}} = C_1 e^{m_1 x} + C_2 e^{m_2 x}$.
- **Real & Equal Roots** ($m_1 = m_2 = m$): $y_{\text{CF}} = (C_1 + C_2 x)e^{mx}$.
- **Complex Conjugate Roots** ($m = \alpha \pm i\beta$): $y_{\text{CF}} = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$.

5.3 Cauchy-Euler Differential Equations

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Cauchy-Euler Substitution

Form: $ax^2 \frac{d^2y}{dx^2} + bx\frac{dy}{dx} + cy = f(x)$. **Substitution**: Set $x = e^z \implies z = \ln x$. **Operators**: $x\frac{dy}{dx} = Dy$, $x^2\frac{d^2y}{dx^2} = D(D-1)y$ where $D = \frac{d}{dz}$. This converts the equation into a standard linear ODE with constant coefficients w.r.t z!

5.4 Legendre Polynomials & Series Solutions

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Legendre Differential Equation Properties

Legendre Equation: $(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + n(n+1)y = 0$. **Rodrigues' Formula for $P_n(x)$**:

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n$$

First Few Polynomials: $P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{1}{2}(3x^2-1), \quad P_3(x) = \frac{1}{2}(5x^3 - 3x)$.
Orthogonality Property:

$$\int_{-1}^{1} P_m(x)P_n(x)\,dx = \begin{cases} 0 & \text{if } m \neq n \\ \frac{2}{2n+1} & \text{if } m = n \end{cases}$$

5.5 Partial Differential Equations (First-Order Non-Linear)

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Lagrange's Method for First-Order Quasi-Linear PDEs

Standard Form: $P(x,y,z)p + Q(x,y,z)q = R(x,y,z)$ where $p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}$. **Lagrange's Auxiliary Equations**:

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

Solve the auxiliary ratios to find two independent integrals $u(x,y,z) = c_1$ and $v(x,y,z) = c_2$. The general solution is $F(u,v) = 0$ or $u = f(v)$.

6 Unit VI: CASIO fx-991ES PLUS Calculator Companion & Exam Strategy

6.1 Step-by-Step Calculator Operations

EASY EXAM TRICK / DESI PRO-TIP: Radian Mode Calculator Setup

Always set to **Radian Mode** before starting your calculus exam: Press **SHIFT** → **MODE** → **4: Rad** (Check for small **R** on top of screen).

EASY EXAM TRICK / DESI PRO-TIP: Limit Evaluation Trick (CALC Button)

To check limits like $\lim_{x\to 5} \frac{x-5}{x^2-25}$: 1. Type expression using fraction key and **ALPHA** +) for X. 2. Press **CALC**. 3. Enter $X = 5.0001$. Press **=**. Output shows $0.09999 \approx \mathbf{0.1} = \mathbf{1/10}$.

EASY EXAM TRICK / DESI PRO-TIP: Derivative Verification (d/dx Button)

To check derivative of $y = \cos(x^2)$ at $x = 1$: Press **SHIFT** + $\int dx$ → Type $\cos(X^2)$ → Set $X = 1$ → Press **=** → Output: -1.6829.

6.2 Final Exam Execution Blueprint

3-Hour 60-Mark SEE Exam Action Strategy

1. **First 45 Mins — Section C (30 Marks)**: Attack the 3 heavy-hitter 10-mark problems first (Ellipse Area, Polar Circles, Triple Integral / Volume of Revolution).
2. **Next 60 Mins — Section B (20 Marks)**: Solve 4 out of 6 questions (Gradient $\nabla\Phi$, Tangent/Normal, 2-Variable Extrema).
3. **Next 30 Mins — Section A (10 Marks)**: Complete short answer questions and verify numerical limits via calculator.
4. **Final 45 Mins**: Double-check calculations, verify units, and review answer presentation.

7 Extended Practice Bank: Step-by-Step Worked PYQs & Syllabus Illustrations

7.1 Unit I Extended Archive: Limits, Continuity & Partials

IMPORTANT PYQ / EXAM TOPIC: Extended Limit Evaluation Problems (Assignments & Mid-Sem PYQs)

Problem 1 (Assignment Q1): Evaluate $\lim_{x\to 0} \frac{3x^2+2x+1}{5x^2+6x+7}$.

Step-by-Step Solution. 1. Check for indeterminate form by direct substitution: Numerator at $x = 0$: $3(0)^2+2(0)+1 = 1$. Denominator at $x = 0$: $5(0)^2+6(0)+7 = 7$. 2. Since the denominator is non-zero, substitution is valid:

$$\lim_{x\to 0} \frac{3x^2+2x+1}{5x^2+6x+7} = \frac{\mathbf{1}}{\mathbf{7}}$$

□

Problem 2 (Assignment Q2): Evaluate $\lim_{x\to 0} \frac{3x^2+4x}{5x-7x^2}$.

Step-by-Step Solution. 1. Direct substitution gives $\frac{0}{0}$ (Indeterminate form). 2. Factor out x from numerator and denominator:

$$\lim_{x\to 0}\frac{x(3x+4)}{x(5-7x)}=\lim_{x\to 0}\frac{3x+4}{5-7x}$$

3. Substitute $x=0$:

$$\frac{3(0)+4}{5-7(0)}=\frac{\mathbf{4}}{\mathbf{5}}$$

□

Problem 3 (Assignment Q3): Evaluate $\lim_{x\to 1}\frac{x^2+x-2}{x^2-4x+3}$.

Step-by-Step Solution. 1. Direct substitution gives $\frac{1+1-2}{1-4+3}=\frac{0}{0}$. 2. Factorize quadratic expressions: Numerator: $x^2+x-2=(x-1)(x+2)$. Denominator: $x^2-4x+3=(x-1)(x-3)$. 3. Cancel common indeterminate factor $(x-1)$:

$$\lim_{x\to 1}\frac{(x-1)(x+2)}{(x-1)(x-3)}=\lim_{x\to 1}\frac{x+2}{x-3}$$

4. Substitute $x=1$:

$$\frac{1+2}{1-3}=\frac{3}{-2}=-\frac{\mathbf{3}}{\mathbf{2}}$$

□

Problem 4 (Assignment Q4): Evaluate $\lim_{x\to 2}\frac{x^{10}-2^{10}}{x-2}$.

Step-by-Step Solution. 1. Use standard formula: $\lim_{x\to a}\frac{x^n-a^n}{x-a}=n\cdot a^{n-1}$. 2. Here $a=2$ and $n=10$:

$$\text{Limit}=10\cdot 2^{10-1}=10\cdot 2^9=10\cdot 512=\mathbf{5120}$$

□

Problem 5 (Assignment Q8): Evaluate $\lim_{x\to 3}\frac{x^5-243}{x^2-9}$.

Step-by-Step Solution. 1. Rewrite $243=3^5$ and $9=3^2$:

$$\lim_{x\to 3}\frac{x^5-3^5}{x^2-3^2}$$

2. Use identity $\lim_{x\to a}\frac{x^n-a^n}{x^m-a^m}=\frac{n}{m}a^{n-m}$: Here $a=3, n=5, m=2$:

$$\text{Limit}=\frac{5}{2}\cdot 3^{5-2}=\frac{5}{2}\cdot 3^3=\frac{5}{2}\cdot 27=\frac{\mathbf{135}}{\mathbf{2}}$$

□

Problem 6 (Assignment Q6): Evaluate $\lim_{x\to 0}\frac{\sin 5x}{4x}$.

Step-by-Step Solution. 1. Use standard limit $\lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1$. 2. Multiply and divide by 5:

$$\lim_{x\to 0}\left(\frac{5}{4}\cdot\frac{\sin 5x}{5x}\right)=\frac{5}{4}\cdot(1)=\frac{\mathbf{5}}{\mathbf{4}}$$

□

Problem 7 (Assignment Q10): Evaluate $\lim_{\theta\to 0}\frac{\sin 8\theta}{\tan 3\theta}$.

Step-by-Step Solution. 1. Divide numerator and denominator by θ:

$$\lim_{\theta\to 0}\frac{\frac{\sin 8\theta}{\theta}}{\frac{\tan 3\theta}{\theta}}=\frac{8\cdot\lim_{\theta\to 0}\frac{\sin 8\theta}{8\theta}}{3\cdot\lim_{\theta\to 0}\frac{\tan 3\theta}{3\theta}}=\frac{8(1)}{3(1)}=\frac{\mathbf{8}}{\mathbf{3}}$$

□

IMPORTANT PYQ / EXAM TOPIC: Single-Variable Differentiation Archive (Assignment Questions 11–20)

Problem 8 (Assignment Q11): Find $\frac{dy}{dx}$ for:

1. $y = \frac{1}{x^2} = x^{-2} \implies \frac{dy}{dx} = -2x^{-3} = -\mathbf{\frac{2}{x^3}}$.
2. $y = x^3 + 2 \implies \frac{dy}{dx} = \mathbf{3x^2}$.
3. $y = \sqrt{x} = x^{1/2} \implies \frac{dy}{dx} = \frac{1}{2}x^{-1/2} = \mathbf{\frac{1}{2\sqrt{x}}}$.
4. $y = \frac{1}{\sin x} = \csc x \implies \frac{dy}{dx} = -\csc \mathbf{x} \cot \mathbf{x}$.

Problem 9 (Assignment Q13): Differentiate $y = e^x \sin x$.

Solution. Apply Product Rule $\frac{d}{dx}(u \cdot v) = u'v + uv'$:

$$\frac{dy}{dx} = (e^x)' \sin x + e^x (\sin x)' = e^x \sin x + e^x \cos x = \mathbf{e^x}(\sin \mathbf{x} + \cos \mathbf{x})$$

□

Problem 10 (Assignment Q14): Differentiate $y = x^2 e^x \sin x$.

Solution. Apply 3-factor Product Rule:

$$\frac{dy}{dx} = (x^2)' e^x \sin x + x^2 (e^x)' \sin x + x^2 e^x (\sin x)' = 2xe^x \sin x + x^2 e^x \sin x + x^2 e^x \cos x = \mathbf{xe^x}(\mathbf{2} \sin \mathbf{x} + \mathbf{x} \sin \mathbf{x} + \mathbf{x} \cos \mathbf{x}$$

□

Problem 11 (Assignment Q16): Differentiate $y = \frac{1+x+x^2}{1-x+x^2}$.

Solution. Apply Quotient Rule $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$: Let $u = 1 + x + x^2 \implies u' = 1 + 2x$; $v = 1 - x + x^2 \implies v' = -1 + 2x$.

$$\frac{dy}{dx} = \frac{(1+2x)(1-x+x^2) - (1+x+x^2)(-1+2x)}{(1-x+x^2)^2} = \frac{\mathbf{2(1-x^2)}}{\mathbf{(1-x+x^2)^2}}$$

□

Problem 12 (Assignment Q18): Chain Rule Problems:

1. $y = (2x^2 - 3x + 1)^3 \implies \frac{dy}{dx} = 3(2x^2 - 3x + 1)^2 \cdot (4x - 3) = \mathbf{(12x - 9)(2x^2 - 3x + 1)^2}$.
2. $y = \cos(e^{5x}) \implies \frac{dy}{dx} = -\sin(e^{5x}) \cdot e^{5x} \cdot 5 = \mathbf{-5e^{5x}} \sin(\mathbf{e^{5x}})$.
3. $y = \ln(\sec x + \tan x) \implies \frac{dy}{dx} = \frac{1}{\sec x + \tan x} \cdot (\sec x \tan x + \sec^2 x) = \frac{\sec x(\tan x + \sec x)}{\sec x + \tan x} = \sec \mathbf{x}$.

IMPORTANT PYQ / EXAM TOPIC: Multivariable Partials Euler Proof Archive (Assignment Questions 21–28)

Problem 13 (Assignment Q23): Find $\frac{\partial^2 u}{\partial x^2}$ and $\frac{\partial^2 u}{\partial y^2}$ for $u = x^3 \tan y$.

Solution. 1. $\frac{\partial u}{\partial x} = 3x^2 \tan y \implies \frac{\partial^2 u}{\partial x^2} = \mathbf{6x} \tan \mathbf{y}$. 2. $\frac{\partial u}{\partial y} = x^3 \sec^2 y \implies \frac{\partial^2 u}{\partial y^2} = x^3(2 \sec y \cdot \sec y \tan y) = \mathbf{2x^3} \sec^2 \mathbf{y} \tan \mathbf{y}$. □

Problem 14 (Assignment Q25): If $u = \ln(x^3 + y^3)$, find $\frac{\partial^2 u}{\partial x^2}$.

Solution. 1. $\frac{\partial u}{\partial x} = \frac{3x^2}{x^3+y^3}$. 2. Apply Quotient Rule w.r.t x:

$$\frac{\partial^2 u}{\partial x^2} = \frac{(6x)(x^3+y^3) - (3x^2)(3x^2)}{(x^3+y^3)^2} = \frac{6x^4 + 6xy^3 - 9x^4}{(x^3+y^3)^2} = \frac{\mathbf{3x(2y^3 - x^3)}}{\mathbf{(x^3+y^3)^2}}$$

□

Problem 15 (Assignment Q27): If $u = x^3 + 3x^2y + 3xy^2 + y^3$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3u$.

Solution. 1. Check homogeneity: $u(tx, ty) = t^3x^3 + 3t^3x^2y + 3t^3xy^2 + t^3y^3 = t^3u(x, y)$. 2. u is homogeneous of degree $n = 3$. 3. By Euler's Theorem: $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu = \mathbf{3u}$ (Hence Proved!). □

7.2 Unit IV Extended Archive: Polar Areas & Multiple Integral Proofs

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Cardioid Area Calculation by Double Integration

Problem Statement: Find the area enclosed by the cardioid $r = a(1 + \cos\theta)$.

Step-by-Step Solution. 1. Use symmetry about the initial line $\theta = 0$: Total Area $A = 2 \times \int_0^{\pi} \int_0^{a(1+\cos\theta)} r\, dr\, d\theta$. 2. Inner integral w.r.t r:

$$\int_0^{a(1+\cos\theta)} r\, dr = \left[\frac{r^2}{2}\right]_0^{a(1+\cos\theta)} = \frac{a^2}{2}(1+\cos\theta)^2$$

3. Outer integral w.r.t θ:

$$A = 2 \cdot \frac{a^2}{2} \int_0^{\pi} (1 + 2\cos\theta + \cos^2\theta)\, d\theta = a^2 \int_0^{\pi} \left(1 + 2\cos\theta + \frac{1+\cos 2\theta}{2}\right) d\theta$$

$$A = a^2 \int_0^{\pi} \left(\frac{3}{2} + 2\cos\theta + \frac{1}{2}\cos 2\theta\right) d\theta = a^2 \left[\frac{3}{2}\theta + 2\sin\theta + \frac{\sin 2\theta}{4}\right]_0^{\pi} = a^2\left(\frac{3\pi}{2}\right) = \mathbf{\frac{3}{2}\pi a^2}$$

□

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Change of Order of Integration: Complete Recipe

Problem Statement: Change the order of integration for $\int_0^1 \int_x^1 f(x, y)\, dy\, dx$.

Step-by-Step Solution. 1. Identify current region R: Vertical strip bounds y from line $y = x$ to horizontal line $y = 1$, while x goes from 0 to 1. 2. Sketch region: Triangle bounded by $x = 0$, $y = 1$, and $y = x$. 3. Change to Horizontal Strips: For a fixed y, x varies from $x = 0$ to $x = y$. The range for y is from 0 to 1. 4. Rewritten Integral:

$$\int_{\mathbf{0}}^{\mathbf{1}} \int_{\mathbf{0}}^{\mathbf{y}} \mathbf{f(x, y)\, dx\, dy}$$

□

7.3 Unit V Extended Archive: Advanced Differential Equations

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Method of Variation of Parameters for 2nd-Order LDEs

Standard Form: $y'' + Py' + Qy = R(x)$. **Algorithm**: 1. Find complementary function $y_{\text{CF}} = c_1 y_1(x) + c_2 y_2(x)$. 2. Calculate Wronskian: $W(y_1, y_2) = y_1 y_2' - y_2 y_1'$. 3. Particular integral is $y_{\text{PI}} = u(x)y_1(x) + v(x)y_2(x)$ where:

$$u(x) = -\int \frac{y_2(x)R(x)}{W}\,dx, \quad v(x) = \int \frac{y_1(x)R(x)}{W}\,dx$$

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Non-Linear First-Order PDEs Charpit's Method

Standard Charpit's Auxiliary Equations: For non-linear PDE $f(x, y, z, p, q) = 0$:

$$\frac{dp}{f_x + pf_z} = \frac{dq}{f_y + qf_z} = \frac{dz}{-pf_p - qf_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

Solve for p and q, substitute into $dz = pdx + qdy$, and integrate to find complete integral.

7.4 Unit V Extended Archive: Comprehensive Differential Equations Problem Bank

IMPORTANT PYQ / EXAM TOPIC: First-Order Linear ODEs Worked Archive

Problem 1: Solve $\frac{dy}{dx} + y\tan x = \sec x$.

Step-by-Step Solution. 1. Identify $P(x) = \tan x$ and $Q(x) = \sec x$. 2. Calculate Integrating Factor (IF):

$$\text{IF} = e^{\int \tan x\, dx} = e^{\ln(\sec x)} = \sec x$$

3. Write general solution formula: $y \cdot (\text{IF}) = \int Q(x) \cdot (\text{IF})\, dx + C$. 4. Substitute:

$$y\sec x = \int \sec x \cdot \sec x\, dx + C = \int \sec^2 x\, dx + C = \tan x + C$$

5. Divide by $\sec x$:

$$\mathbf{y = \sin x + C\cos x}$$

□

Problem 2: Solve $(x^2+1)\frac{dy}{dx} + 2xy = 4x^2$.

Step-by-Step Solution. 1. Standardize by dividing by (x^2+1):

$$\frac{dy}{dx} + \frac{2x}{x^2+1}y = \frac{4x^2}{x^2+1}$$

2. Identify $P(x) = \frac{2x}{x^2+1}$ and $Q(x) = \frac{4x^2}{x^2+1}$. 3. Calculate IF:

$$\text{IF} = e^{\int \frac{2x}{x^2+1}dx} = e^{\ln(x^2+1)} = x^2+1$$

4. General solution:

$$y(x^2+1) = \int \frac{4x^2}{x^2+1}(x^2+1)\, dx + C = \int 4x^2\, dx + C = \frac{4x^3}{3} + C$$

$$\mathbf{y = \frac{4x^3}{3(x^2+1)} + \frac{C}{x^2+1}}$$

□

IMPORTANT PYQ / EXAM TOPIC: Second-Order ODEs with Constant Coefficients Worked Archive

Problem 3: Solve $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{4x}$.

Step-by-Step Solution. 1. Auxiliary Equation: $m^2 - 5m + 6 = 0 \implies (m-2)(m-3) = 0 \implies m_1 = 2, m_2 = 3$. 2. Complementary Function: $y_{\text{CF}} = C_1e^{2x} + C_2e^{3x}$. 3. Particular Integral y_{PI}:

$$y_{\text{PI}} = \frac{1}{D^2 - 5D + 6}e^{4x}$$

Substitute $D = 4$:

$$y_{\text{PI}} = \frac{1}{4^2 - 5(4) + 6}e^{4x} = \frac{1}{16 - 20 + 6}e^{4x} = \frac{e^{4x}}{2}$$

4. General Solution:

$$\mathbf{y = C_1e^{2x} + C_2e^{3x} + \frac{e^{4x}}{2}}$$

□

Problem 4: Solve $\frac{d^2y}{dx^2} + 4y = \sin 3x$.

Step-by-Step Solution. 1. Auxiliary Equation: $m^2+4=0 \implies m=\pm 2i$ ($\alpha=0,\beta=2$). 2. Complementary Function: $y_{\text{CF}}=C_1\cos 2x+C_2\sin 2x$. 3. Particular Integral y_{PI}:

$$y_{\text{PI}}=\frac{1}{D^2+4}\sin 3x$$

Substitute $D^2=-3^2=-9$:

$$y_{\text{PI}}=\frac{1}{-9+4}\sin 3x=-\frac{\sin 3x}{5}$$

4. General Solution:

$$\mathbf{y=C_1\cos 2x+C_2\sin 2x-\frac{\sin 3x}{5}}$$

□

IMPORTANT PYQ / EXAM TOPIC: Cauchy-Euler Substitution Worked Archive

Problem 5: Solve $x^2\frac{d^2y}{dx^2}-3x\frac{dy}{dx}+4y=0$.

Step-by-Step Solution. 1. Substitute $x=e^z \implies z=\ln x$. 2. Replace operators: $x\frac{dy}{dx}=Dy$ and $x^2\frac{d^2y}{dx^2}=D(D-1)y$ where $D=\frac{d}{dz}$. 3. Substitute into ODE:

$$[D(D-1)-3D+4]y=0 \implies (D^2-4D+4)y=0$$

4. Auxiliary equation in m: $m^2-4m+4=0 \implies (m-2)^2=0 \implies m_1=2, m_2=2$ (Equal roots!). 5. Solution in variable z:

$$y(z)=(C_1+C_2z)e^{2z}$$

6. Convert back to x using $z=\ln x$ and $e^{2z}=x^2$:

$$\mathbf{y(x)=(C_1+C_2\ln x)x^2}$$

□

IMPORTANT PYQ / EXAM TOPIC: Lagrange PDE Auxiliary Method Worked Archive

Problem 6: Solve $xp+yq=z$.

Step-by-Step Solution. 1. Identify $P=x$, $Q=y$, $R=z$. 2. Write Lagrange Auxiliary Equations:

$$\frac{dx}{x}=\frac{dy}{y}=\frac{dz}{z}$$

3. Take first ratio pair $\frac{dx}{x}=\frac{dy}{y}$: Integrate: $\ln x=\ln y+\ln c_1 \implies \frac{x}{y}=c_1=u(x,y,z)$. 4. Take second ratio pair $\frac{dy}{y}=\frac{dz}{z}$: Integrate: $\ln y=\ln z+\ln c_2 \implies \frac{y}{z}=c_2=v(x,y,z)$. 5. Write General Solution:

$$\phi\left(\frac{\mathbf{x}}{\mathbf{y}},\frac{\mathbf{y}}{\mathbf{z}}\right)=\mathbf{0} \quad \text{or} \quad \frac{\mathbf{x}}{\mathbf{y}}=\mathbf{f}\left(\frac{\mathbf{y}}{\mathbf{z}}\right)$$

□

7.5 Unit IV Extended Archive: Complete Proofs Vector Integral Theorems

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Green's Theorem Verification Example

Problem Statement: Verify Green's Theorem in the plane for $\oint_C (x^2dy - y^2dx)$ where C is the boundary of the square $[0,1] \times [0,1]$.

Step-by-Step Solution. 1. Identify $P = -y^2$ and $Q = x^2$. 2. Compute partial derivatives:

$$\frac{\partial Q}{\partial x} = 2x, \quad \frac{\partial P}{\partial y} = -2y \implies \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2x - (-2y) = 2(x+y)$$

3. Evaluate Double Integral over region $R = [0,1] \times [0,1]$:

$$\iint_R 2(x+y)\,dx\,dy = 2\int_0^1 \int_0^1 (x+y)\,dx\,dy = 2\int_0^1 \left[\frac{x^2}{2} + xy\right]_0^1 dy = 2\int_0^1 \left(\frac{1}{2} + y\right) dy$$

$$= 2\left[\frac{1}{2}y + \frac{y^2}{2}\right]_0^1 = 2\left(\frac{1}{2} + \frac{1}{2}\right) = \mathbf{2}$$

4. Evaluate Line Integral around 4 sides of the square C: - Side 1 ($y = 0, x \in [0,1], dy = 0$): $\int_0^1 0 = 0$. - Side 2 ($x = 1, y \in [0,1], dx = 0$): $\int_0^1 1^2 dy = 1$. - Side 3 ($y = 1, x \in [1,0], dy = 0$): $\int_1^0 -1^2 dx = 1$. - Side 4 ($x = 0, y \in [1,0], dx = 0$): $\int_1^0 0 = 0$. 5. Sum of Line Integrals: $0+1+1+0 = \mathbf{2}$. 6. Since Line Integral = Double Integral = 2, **Green's Theorem is Verified!** □

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Gauss's Divergence Theorem Verification Example

Problem Statement: Compute $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$ where $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ over the surface of the sphere $x^2 + y^2 + z^2 = a^2$.

Step-by-Step Solution. 1. Use Gauss's Divergence Theorem: $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS = \iiint_V (\nabla \cdot \mathbf{F})\, dV$. 2. Compute Divergence of $\mathbf{F}$:

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$$

3. Evaluate Volume Integral:

$$\iiint_V 3\, dV = 3 \times \text{Volume of sphere of radius } a = 3 \times \left(\frac{4}{3}\pi a^3\right) = \mathbf{4\pi a^3}$$

□

7.6 Unit II Extended Archive: Taylor Series Power Series Applications

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Taylor Series Expansion Around $x = a$

Formula: Taylor series of $f(x)$ about $x = a$:

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \dots$$

Worked Example: Expand $f(x) = \sin x$ in powers of $\left(x - \frac{\pi}{2}\right)$.

Step-by-Step Solution. 1. Let $a = \frac{\pi}{2}$. Compute derivatives at $x = \frac{\pi}{2}$:

$$\begin{aligned} f(x) &= \sin x \implies f(\pi/2) = 1 \\ f'(x) &= \cos x \implies f'(\pi/2) = 0 \\ f''(x) &= -\sin x \implies f''(\pi/2) = -1 \\ f'''(x) &= -\cos x \implies f'''(\pi/2) = 0 \\ f^{(4)}(x) &= \sin x \implies f^{(4)}(\pi/2) = 1 \end{aligned}$$

2. Substitute into Taylor expansion formula:

$$\sin x = 1 + 0\left(x - \frac{\pi}{2}\right) - \frac{1}{2!}\left(x - \frac{\pi}{2}\right)^2 + 0 - \frac{1}{4!}\left(x - \frac{\pi}{2}\right)^4 + \dots$$

$$\sin \mathbf{x} = \mathbf{1} - \frac{\mathbf{1}}{\mathbf{2!}}\left(\mathbf{x} - \frac{\pi}{\mathbf{2}}\right)^{\mathbf{2}} + \frac{\mathbf{1}}{\mathbf{4!}}\left(\mathbf{x} - \frac{\pi}{\mathbf{2}}\right)^{\mathbf{4}} - \dots$$

□

7.7 Comprehensive Solved Practice Set (Units I–V)

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Unit I: Tangent Planes Normal Lines

Problem Statement: Find the equation of the tangent plane and normal line to the surface $z = x^2 + y^2$ at the point $(1, 2, 5)$.

Step-by-Step Solution. 1. Let $F(x, y, z) = x^2 + y^2 - z = 0$. 2. Compute partial derivatives for the normal vector $\mathbf{N} = \nabla F$:

$$\frac{\partial F}{\partial x} = 2x, \quad \frac{\partial F}{\partial y} = 2y, \quad \frac{\partial F}{\partial z} = -1$$

3. Evaluate ∇F at point $(1, 2, 5)$:

$$\mathbf{N} = \nabla F(1, 2, 5) = 2(1)\mathbf{i} + 2(2)\mathbf{j} - 1\mathbf{k} = 2\mathbf{i} + 4\mathbf{j} - \mathbf{k}$$

4. Tangent Plane Equation: $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$:

$$2(x-1) + 4(y-2) - 1(z-5) = 0 \implies 2x - 2 + 4y - 8 - z + 5 = 0 \implies \mathbf{2x + 4y - z = 5}$$

5. Normal Line Equations (Symmetric Form):

$$\frac{\mathbf{x - 1}}{\mathbf{2}} = \frac{\mathbf{y - 2}}{\mathbf{4}} = \frac{\mathbf{z - 5}}{\mathbf{-1}}$$

□

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Unit IV: Double Integral Area of a Parabola Line

Problem Statement: Find the area enclosed by $y = x^2$ and $y = 2x + 3$.

Step-by-Step Solution. 1. Find intersection points: $x^2 = 2x + 3 \implies x^2 - 2x - 3 = 0 \implies (x - 3)(x + 1) = 0 \implies x = -1, x = 3$. 2. Double integral for area:

$$A = \int_{-1}^{3} \left(\int_{x^2}^{2x+3} dy \right) dx = \int_{-1}^{3} (2x + 3 - x^2)\, dx$$

3. Integrate w.r.t x:

$$A = \left[x^2 + 3x - \frac{x^3}{3}\right]_{-1}^{3} = (9 + 9 - 9) - \left(1 - 3 + \frac{1}{3}\right) = 9 - \left(-\frac{5}{3}\right) = 9 + \frac{5}{3} = \mathbf{\frac{32}{3} \approx 10.67}$$

□

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Unit V: Power Series Solution of Differential Equations

Problem Statement: Find the power series solution of $y' - y = 0$ about $x = 0$.

Step-by-Step Solution. 1. Assume power series solution $y = \sum_{n=0}^{\infty} a_n x^n$. 2. Derivative: $y' = \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$. 3. Substitute into $y' - y = 0$:

$$\sum_{n=0}^{\infty} \left[(n+1)a_{n+1} - a_n\right] x^n = 0$$

4. Recurrence Relation: $(n+1)a_{n+1} - a_n = 0 \implies a_{n+1} = \frac{a_n}{n+1}$. 5. Calculate coefficients: $a_1 = a_0$, $a_2 = \frac{a_1}{2} = \frac{a_0}{2!}$, $a_3 = \frac{a_2}{3} = \frac{a_0}{3!}$, $\ldots$, $a_n = \frac{a_0}{n!}$. 6. Solution:

$$y(x) = a_0 \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\right) = \mathbf{a_0 e^x}$$

□

7.8 Unit V Extended Archive: Non-Linear First-Order PDEs Charpit's Standard Forms

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Standard Type 1: $f(p,q) = 0$ (Only p and q present)

Algorithm: 1. Since only p and q are present, assume solution $z = ax + by + c$. 2. Substitute $p = a$ and $q = b$ into $f(a,b) = 0$ and solve for $b = g(a)$. 3. Substitute b back into the solution: $z = ax + g(a)y + c$.
Worked Example: Solve $p^2 + q^2 = 1$.

Step-by-Step Solution. 1. Let $z = ax + by + c \implies p = a, q = b$. 2. Substitute: $a^2 + b^2 = 1 \implies b = \sqrt{1 - a^2}$. 3. Complete Integral: $\mathbf{z = ax + \sqrt{1 - a^2}y + c}$. □

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Standard Type 2: $z = px + qy + f(p,q)$ (Clairaut's Form)

Algorithm: 1. Replace p by constant a and q by constant b. 2. Complete Integral: $z = ax + by + f(a,b)$.
Worked Example: Solve $z = px + qy + p^2 + q^2$.

Step-by-Step Solution. Replace $p \to a$ and $q \to b$:

$$\mathbf{z = ax + by + a^2 + b^2}$$

□

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Standard Type 3: $f(z,p,q)=0$ (Only z,p,q present)

Algorithm: 1. Set z as a function of $u=x+ay \implies p=\frac{dz}{du}, q=a\frac{dz}{du}$. 2. Substitute into $f(z,p,q)=0$ to get a single ODE in z and u. 3. Separate variables and integrate.
Worked Example: Solve $z=p^2+q^2$.

Step-by-Step Solution. 1. Let $u=x+ay \implies p=\frac{dz}{du}, q=a\frac{dz}{du}$. 2. Substitute: $z=\left(\frac{dz}{du}\right)^2+a^2\left(\frac{dz}{du}\right)^2=(1+a^2)\left(\frac{dz}{du}\right)^2$. 3. Solve for $\frac{dz}{du}$:

$$\left(\frac{dz}{du}\right)^2=\frac{z}{1+a^2} \implies \frac{dz}{du}=\frac{\sqrt{z}}{\sqrt{1+a^2}}$$

4. Separate variables:

$$\frac{dz}{\sqrt{z}}=\frac{du}{\sqrt{1+a^2}}$$

5. Integrate both sides:

$$2\sqrt{z}=\frac{u}{\sqrt{1+a^2}}+c \implies \mathbf{2\sqrt{z}=\frac{x+ay}{\sqrt{1+a^2}}+c}$$

□

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Standard Type 4: $f_1(x,p)=f_2(y,q)$ (Separable Variables)

Algorithm: 1. Equate both sides to an arbitrary constant k: $f_1(x,p)=k \implies p=g_1(x,k)$ and $f_2(y,q)=k \implies q=g_2(y,k)$. 2. Substitute into $dz=pdx+qdy=g_1(x,k)dx+g_2(y,k)dy$. 3. Integrate to find complete integral $z=\int g_1(x,k)dx+\int g_2(y,k)dy+c$.
Worked Example: Solve $p^2+q^2=x+y$.

Step-by-Step Solution. 1. Separate variables: $p^2-x=y-q^2=k$. 2. $p^2-x=k \implies p=\sqrt{x+k}$. 3. $y-q^2=k \implies q=\sqrt{y-k}$. 4. Substitute into $dz=pdx+qdy$:

$$dz=\sqrt{x+k}\,dx+\sqrt{y-k}\,dy$$

5. Integrate both sides:

$$\mathbf{z=\frac{2}{3}(x+k)^{3/2}+\frac{2}{3}(y-k)^{3/2}+c}$$

□

7.9 Unit I Extended Archive: Directional Derivative Gradient Applications

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Maximum Rate of Change Normal Vectors

Theorem: For a scalar function $f(x,y,z)$:

1. The **maximum rate of change** of f at a point occurs in the direction of the gradient vector ∇f.
2. The **maximum magnitude** of this rate of change is equal to the length of the gradient vector, $|\nabla f|$.
3. The gradient vector ∇f is always **perpendicular (normal)** to the level surface $f(x,y,z)=C$.

Worked Problem: Find the maximum rate of change of $\Phi(x,y,z)=x^2yz^3$ at the point $(1,3,2)$

and the direction in which it occurs.

Step-by-Step Solution. 1. Compute partial derivatives of Φ:

$$\frac{\partial \Phi}{\partial x} = 2xyz^3, \quad \frac{\partial \Phi}{\partial y} = x^2z^3, \quad \frac{\partial \Phi}{\partial z} = 3x^2yz^2$$

2. Evaluate $\nabla\Phi$ at point $(1, 3, 2)$:

$$\left.\frac{\partial \Phi}{\partial x}\right|_{(1,3,2)} = 2(1)(3)(2^3) = 6(8) = 48$$
$$\left.\frac{\partial \Phi}{\partial y}\right|_{(1,3,2)} = (1)^2(2^3) = 8$$
$$\left.\frac{\partial \Phi}{\partial z}\right|_{(1,3,2)} = 3(1)^2(3)(2^2) = 9(4) = 36$$

So $\nabla\boldsymbol{\Phi}(\mathbf{1}, \mathbf{3}, \mathbf{2}) = \mathbf{48i} + \mathbf{8j} + \mathbf{36k}$. 3. The direction in which the maximum rate of change occurs is along the vector $\mathbf{48i} + \mathbf{8j} + \mathbf{36k}$. 4. The maximum magnitude of rate of change is $|\nabla\Phi|$:

$$|\nabla\Phi| = \sqrt{48^2 + 8^2 + 36^2} = \sqrt{2304 + 64 + 1296} = \sqrt{3664} = \mathbf{4\sqrt{229} \approx 60.53}$$

□

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Angle Between Two Intersecting Surfaces

Formula: The acute angle θ between two surfaces $f(x, y, z) = c_1$ and $g(x, y, z) = c_2$ at an intersection point is given by the angle between their normal vectors $\mathbf{N}_1 = \nabla f$ and $\mathbf{N}_2 = \nabla g$:

$$\cos\theta = \frac{|\nabla f \cdot \nabla g|}{|\nabla f||\nabla g|}$$

Worked Problem: Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$.

Step-by-Step Solution. 1. Let $f(x, y, z) = x^2 + y^2 + z^2 - 9 = 0 \implies \nabla f = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$. At point $(2, -1, 2)$:

$$\mathbf{N}_1 = \nabla f(2, -1, 2) = 4\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$$

Magnitude $|\mathbf{N}_1| = \sqrt{16 + 4 + 16} = \sqrt{36} = 6$. 2. Let $g(x, y, z) = x^2 + y^2 - z - 3 = 0 \implies \nabla g = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}$. At point $(2, -1, 2)$:

$$\mathbf{N}_2 = \nabla g(2, -1, 2) = 4\mathbf{i} - 2\mathbf{j} - \mathbf{k}$$

Magnitude $|\mathbf{N}_2| = \sqrt{16 + 4 + 1} = \sqrt{21}$. 3. Compute Dot Product $\mathbf{N}_1 \cdot \mathbf{N}_2$:

$$\mathbf{N}_1 \cdot \mathbf{N}_2 = (4)(4) + (-2)(-2) + (4)(-1) = 16 + 4 - 4 = 16$$

4. Calculate angle θ:

$$\cos\theta = \frac{16}{6\sqrt{21}} = \frac{8}{3\sqrt{21}} \implies \theta = \arccos\left(\frac{\mathbf{8}}{\mathbf{3\sqrt{21}}}\right) \approx \mathbf{54.4^\circ}$$

□

7.10 Unit IV Extended Archive: Divergence Curl Solved Problem Set

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Solenoidal Irrotational Vector Fields

Problem 1: Find the constant a such that the vector field $\mathbf{F} = (x+3y)\mathbf{i} + (y-2z)\mathbf{j} + (x+az)\mathbf{k}$ is solenoidal.

Step-by-Step Solution. 1. Definition of Solenoidal: $\nabla \cdot \mathbf{F} = 0$. 2. Compute partial derivatives:

$$\frac{\partial}{\partial x}(x+3y) = 1, \quad \frac{\partial}{\partial y}(y-2z) = 1, \quad \frac{\partial}{\partial z}(x+az) = a$$

3. Sum partials to find divergence:

$$\nabla \cdot \mathbf{F} = 1 + 1 + a = 2 + a$$

4. Equate to 0:

$$2 + a = 0 \implies \mathbf{a = -2}$$

□

Problem 2: Show that the vector field $\mathbf{F} = (y^2 \cos x + z^3)\mathbf{i} + (2y \sin x - 4)\mathbf{j} + (3xz^2 + 2)\mathbf{k}$ is irrotational, and find its scalar potential ϕ.

Step-by-Step Solution. 1. Definition of Irrotational: $\nabla \times \mathbf{F} = \mathbf{0}$. 2. Compute Curl $\mathbf{F}$:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 \cos x + z^3 & 2y \sin x - 4 & 3xz^2 + 2 \end{vmatrix}$$

- **i**-component: $\frac{\partial}{\partial y}(3xz^2 + 2) - \frac{\partial}{\partial z}(2y \sin x - 4) = 0 - 0 = 0$. - **j**-component: $\frac{\partial}{\partial z}(y^2 \cos x + z^3) - \frac{\partial}{\partial x}(3xz^2 + 2) = 3z^2 - 3z^2 = 0$. - **k**-component: $\frac{\partial}{\partial x}(2y \sin x - 4) - \frac{\partial}{\partial y}(y^2 \cos x + z^3) = 2y \cos x - 2y \cos x = 0$. Thus $\nabla \times \mathbf{F} = \mathbf{0}$, so $\mathbf{F}$ is **Irrotational**. 3. Find Scalar Potential ϕ such that $\mathbf{F} = \nabla\phi$:

$$\frac{\partial \phi}{\partial x} = y^2 \cos x + z^3 \implies \phi = y^2 \sin x + xz^3 + g(y,z)$$
$$\frac{\partial \phi}{\partial y} = 2y \sin x - 4 \implies \phi = y^2 \sin x - 4y + h(x,z)$$
$$\frac{\partial \phi}{\partial z} = 3xz^2 + 2 \implies \phi = xz^3 + 2z + k(x,y)$$

4. Combining all terms gives:

$$\phi(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \mathbf{y^2} \sin \mathbf{x} + \mathbf{xz^3} - \mathbf{4y} + \mathbf{2z} + \mathbf{C}$$

□

7.11 Unit V Extended Archive: Legendre Polynomial Orthogonality Recurrence Proofs

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Legendre Polynomial Recurrence Relations

Important Recurrence Identities:

1. $(2n+1)xP_n(x) = (n+1)P_{n+1}(x) + nP_{n-1}(x)$.
2. $nP_n(x) = xP_n'(x) - P_{n-1}'(x)$.

3. $(2n+1)P_n(x) = P'_{n+1}(x) - P'_{n-1}(x)$.

Worked Problem: Express $f(x) = x^3 + 2x^2 - x + 1$ in terms of Legendre polynomials.

Step-by-Step Solution. 1. Standard Legendre Polynomials: $P_0(x) = 1$, $P_1(x) = x$, $P_2(x) = \frac{1}{2}(3x^2 - 1) \implies x^2 = \frac{2P_2(x)+1}{3} = \frac{2}{3}P_2(x) + \frac{1}{3}P_0(x)$. $P_3(x) = \frac{1}{2}(5x^3 - 3x) \implies x^3 = \frac{2P_3(x)+3x}{5} = \frac{2}{5}P_3(x) + \frac{3}{5}P_1(x)$. 2. Substitute $x^3, x^2, x, 1$ into $f(x)$:

$$f(x) = \left(\frac{2}{5}P_3(x) + \frac{3}{5}P_1(x)\right) + 2\left(\frac{2}{3}P_2(x) + \frac{1}{3}P_0(x)\right) - P_1(x) + P_0(x)$$

3. Combine coefficients:

$$f(x) = \frac{2}{5}P_3(x) + \frac{4}{3}P_2(x) + \left(\frac{3}{5} - 1\right)P_1(x) + \left(\frac{2}{3} + 1\right)P_0(x)$$

$$\mathbf{f(x) = \frac{2}{5}P_3(x) + \frac{4}{3}P_2(x) - \frac{2}{5}P_1(x) + \frac{5}{3}P_0(x)}$$

□

7.12 Unit V Extended Archive: System of Differential Equations Modeling Applications

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Simultaneous Linear Differential Equations

Problem Statement: Solve the system of simultaneous differential equations:

$$\frac{dx}{dt} + y = 0, \quad \frac{dy}{dt} - x = 0$$

given initial conditions $x(0) = 1, y(0) = 0$.

Step-by-Step Solution. 1. Express using operator $D = \frac{d}{dt}$:

$$Dx + y = 0 \implies y = -Dx \quad \text{— (Eq 1)}$$

$$Dy - x = 0 \quad \text{— (Eq 2)}$$

2. Substitute $y = -Dx$ into (Eq 2):

$$D(-Dx) - x = 0 \implies -D^2x - x = 0 \implies (D^2 + 1)x = 0$$

3. Auxiliary equation: $m^2 + 1 = 0 \implies m = \pm i$. 4. General solution for $x(t)$:

$$x(t) = C_1 \cos t + C_2 \sin t$$

5. Find $y(t)$ using $y = -\frac{dx}{dt}$:

$$y(t) = -\frac{d}{dt}(C_1 \cos t + C_2 \sin t) = -(-C_1 \sin t + C_2 \cos t) = C_1 \sin t - C_2 \cos t$$

6. Apply initial conditions $x(0) = 1, y(0) = 0$: - $x(0) = C_1(1) + C_2(0) = 1 \implies C_1 = 1$. - $y(0) = C_1(0) - C_2(1) = 0 \implies C_2 = 0$. 7. Particular Solution:

$$\mathbf{x(t) = \cos t}, \quad \mathbf{y(t) = \sin t}$$

□

7.13 Unit V Extended Archive: Exact Differential Equations Integrating Factors

SYLLABUS TOPIC / CONCEPT & ILLUSTRATION: Exact Differential Equations

Form: $M(x,y)dx + N(x,y)dy = 0$. **Condition for Exactness**:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Solution Recipe:

$$\int_{y=\text{const}} M\,dx + \int (\text{terms in } N \text{ free from } x)\,dy = C$$

Worked Problem: Solve $(2xy+3)dx + (x^2-1)dy = 0$.

Step-by-Step Solution. 1. Identify $M = 2xy+3$ and $N = x^2-1$. 2. Check exactness: $\frac{\partial M}{\partial y} = 2x$, $\frac{\partial N}{\partial x} = 2x$. Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 2x$, the equation is **Exact**! 3. Apply solution formula:

$$\int_{y=\text{const}} (2xy+3)\,dx + \int (-1)\,dy = C$$

$$yx^2 + 3x - y = C \implies \mathbf{x^2y + 3x - y = C}$$

□